

Chapter 5

Two Faces of Geopolitics of Knowledge



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“the sovereignty lieth hid in knowledge”

Francis Bacon

5.1 Introduction: Geopolitics of Knowledge

Knowledge is central to the extant characterizations of our world such as knowledge economy, information society, network society and surveillance capitalism. Geopolitics of knowledge is real and of highest relevance because of its impact on the sustainability, development and peace of its protagonists. The push and pull over knowledge in international relations only corroborates the Baconian insight of “knowledge is power” (Bacon, 1597/1825).

This chapter defines geopolitics as “Politics, especially relations between states, as influenced by geographical factors” (OED, 2020). The English noun ‘geopolitics’ was first attested in the 1901 essay *England at close of Nineteenth Century to The International Monthly* magazine by Emil Reich (1901), probably a translation of the word *geopolitik* coined by Swedish political philosopher Rudolf Kjellén a year earlier (Kost, 1989). The term stayed dormant for decades but emerged during the World War II and, as a second upsurge, in the 1980s (Fig. 5.1).¹

The prefix ‘geo-’ denoting the paramount importance of geography in the politics dates back to the European colonialism. After the 1871 unification of Germany, ‘*Wissen ist macht, geographisches wissen ist weltmacht*’ (Knowledge is power,

¹ The word/phrase frequency figures in this chapter display results from the Google Books Ngram Viewer, which is based on the latest database—the 2019 *corpora* (Ref. Michel et al., 2011).

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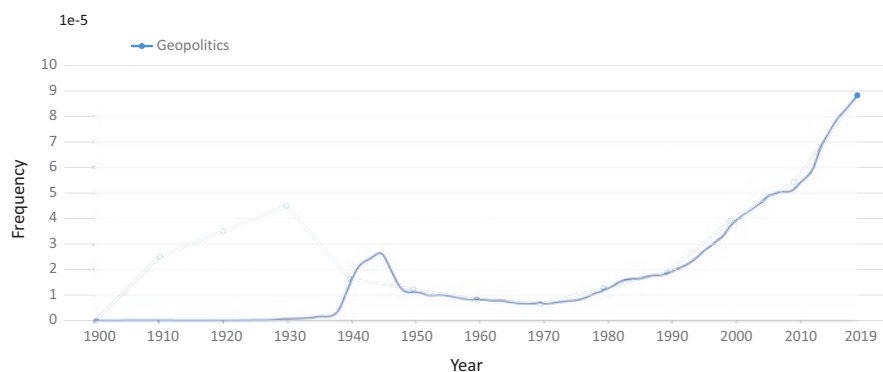


Fig. 5.1 Usage frequency of the word ‘geopolitics’. (Reproduced from Google Books Ngram Viewer, 23 March 2021)

geographical knowledge is world-power) became a slogan in the German geography education and public discourse to foster and support an expansionist and hegemonic politics that would exacerbate during the National Socialist period (Shimazu, 2015). *Geopolitiks* of the German Reich was intellectually justified and nurtured with imperatives of world-power, hence, it “encouraged the development of the political and social sciences” (Kost, 1989, p. 374). While the seventeenth century ‘knowledge is power’ appeared as a divine attribute—*potestas Dei*—in Bacon’s theology, the German ‘*wissen ist macht*’ in the nineteenth century was a geopolitical doctrine.

Today, the recurrent nouns after the ‘geopolitics of *’ are in order of frequency: Knowledge, energy, oil, capitalism, Europe, war, South, Central, information and race (Fig. 5.2). Knowledge is clearly the most written and debated upon topic in contemporary geopolitics from 2010 onwards and it shows a sharper and sustained ascend in frequency of use than the geopolitics over natural resources. The discourse of ‘Geopolitics of Knowledge’ (henceforward, GPK) arose in the 1990s and quickly intensified through the 2000s and 2010s (Fig. 5.3).

The GPK-related discourses are many and their scope, wide. In the outset, this chapter looks into the *problématique* of GPK as is customarily presented in postcolonial studies. The main question arising from it is that the knowledge production is the main analytical perspectives of GPK in postcolonial studies. The central thesis in addressing this question is that the epicenter of GPK from colonialism onward has been the knowledge transfer rather than its production. This claim is illustrated with the ongoing economic neo-imperialism and the Sino-American tensions. The aim and purpose in this study are, thus, to argue that the *problématique* of GPK can be better understood by examining global knowledge transfer and its modalities such as knowledge exchange, gradients of knowledge-power, and the frequently coercive dissemination of knowledge.

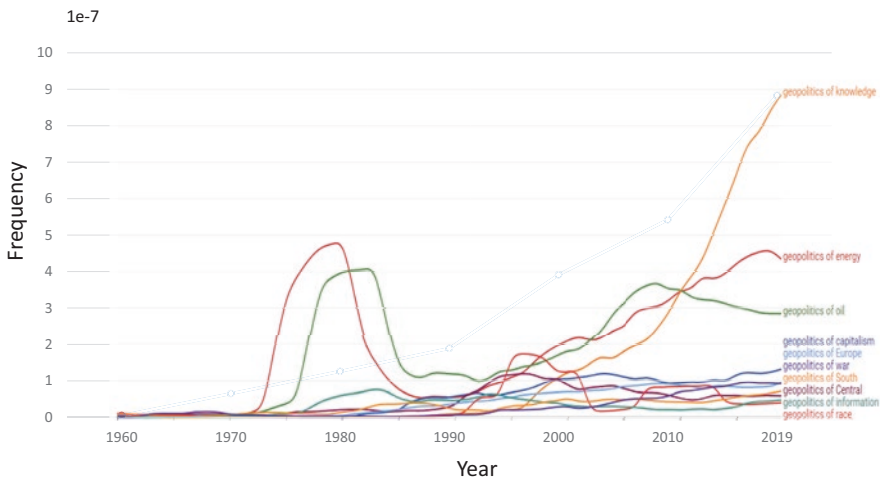


Fig. 5.2 Usage frequency of the nouns after ‘geopolitics of *’. (Reproduced from Google Books Ngram Viewer, 23 March 2021)

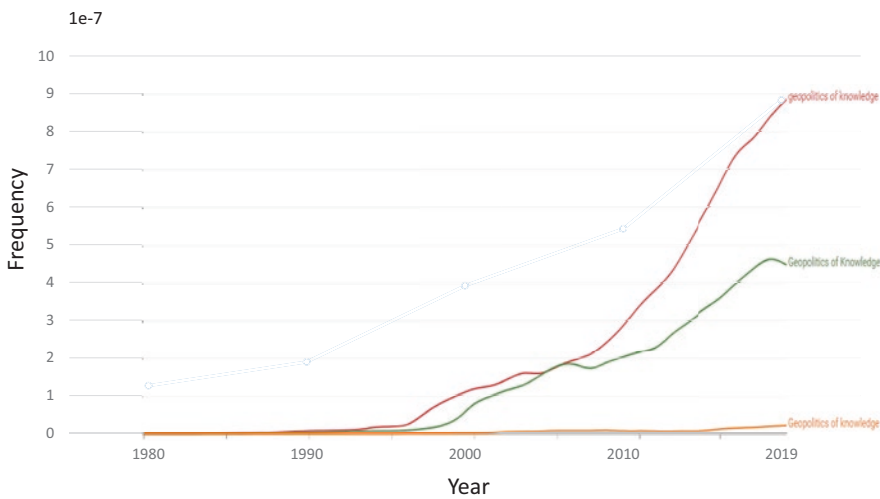


Fig. 5.3 Usage frequency of ‘geopolitics of knowledge’, case-insensitive. (Reproduced from Google Books Ngram Viewer, 23 March 2021)

This chapter devises two analytical steps, namely ‘knowledge production’ and ‘knowledge transfer’. The discussions on the first half of the chapter about knowledge production draw on the extant postcolonial studies. The second half showcases the ongoing Sino-American tensions to demonstrate different modes of knowledge transfer that are crucial to the process of decoding contemporary GPK. The chapter ends with a discussion-conclusion.

5.2 Knowledge Production

A familiar description of the society today would run something like “Paradigm shifts in production, consumption, organization, and the nature of jobs are catalyzed by the progress in technology, Artificial Intelligence and Robotics in particular, leading to the impacts on human lives and driving societal changes.” Similar descriptors can be found in the idea of the *Fourth Industrial Revolution*. Through the periodical he edits, Klaus Schwab promotes among the international economic forum circles the notion of *Fourth Industrial Revolution*, which purportedly has a direct impact on geopolitics (2018). Schwab briefly explains what he means but cannot really define the *Fourth Industrial Revolution* (2015):

There are three reasons why today’s transformations represent not merely a prolongation of the Third Industrial Revolution but rather the arrival of a Fourth and distinct one: velocity, scope, and systems impact. The speed of current breakthroughs has no historical precedent [...], the Fourth is evolving at an exponential rather than a linear pace.

Caution should be called upon any causal extrapolations based on the idea that a lot more of the same is new. Since “revolution is not only incompatible but often actually incommensurable with that which has gone before” (Kuhn, 1970, p. 103), the proposal of the *Fourth Industrial Revolution* based on the amount of knowledge produced is missing out essential characteristics of scientific and technological revolutions—incommensurability if not even a total incompatibility.

5.3 Ownership of Knowledge

The English term ‘knowledge production’ appeared in books only after the 1960s and became popular in the 1980s (Fig. 5.4).

Not entirely by chance, there was an initial international patent law in the same period, the *Patent Cooperation Treaty*, which was ratified by eight states in 1970. A broader international consensus was reached with the *Patent Law Treaty*, which was subscribed by 53 states and the *European Patent Organisation* in the year 2000.

We have to note that international patents are probably one of the most tangible and reliable indicators of global knowledge production although they surely do not and cannot represent the objective quanta and qualia of the entire body of knowledge. We also need to take into account that the dynamics concerning patents are frequently changing because of the variety of items and ways they are measured, for example, the relationship between the number of patent applications versus the number of licenses granted differ substantially. The same holds true if one considers the type of patents such as a *Triadic Patent*—approved by Japan, the US and Europe—and whether it includes all fields or high-tech field only, and to what extent the three major patent offices are used or not.

With international patents as indicators, we can grasp the kind and ownership of potentially profitable knowledge, for example, the actual ownership might belong to

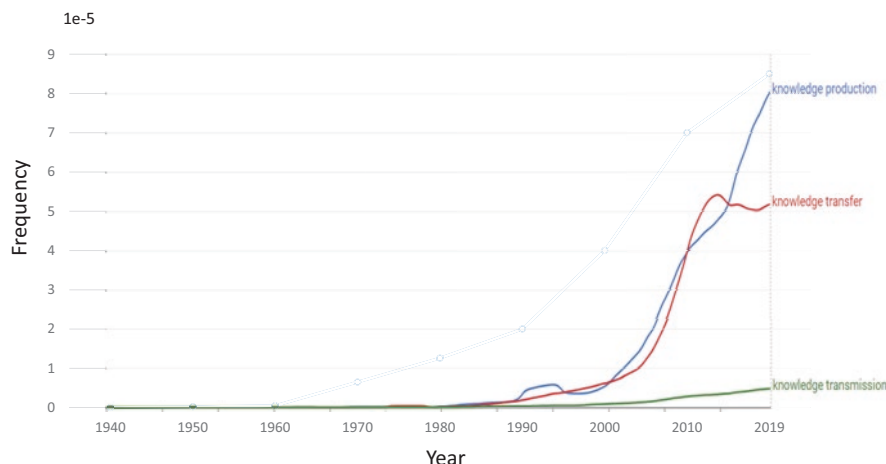


Fig. 5.4 Usage frequency of ‘geopolitics of knowledge’, case-insensitive. (Reproduced from Google Books Ngram Viewer, 23 March 2021)

the patent applicant/holder rather than the actual inventors. Research by Francesco Lissoni (2012) takes full advantage of this phenomenon of the appropriation of knowledge—produced by others—and offers insights into who and where are the real knowledge producers. The role of the European academia and academics in the production of patents is the focus of Lissoni’s research: “academic inventors’ contribution to patenting is concentrated in science-based technologies, such as Pharmaceuticals & Biotechnology (which includes Cosmetics), followed (at some distance) by Chemicals & Materials” (2012, p. 199). Of all patents, only about 3–18% are actually invented by academics. The same research also reveals that European ‘academic patents’ are mostly owned by industry companies (between 60% and 81% depending on the country). The data from the United States display a very different situation with 68.7% of the patents owned by universities whereas companies own 24.2% and individual academics, 5.3%. This phenomenon could partly be explained with decades of incorporation of American universities and/or the research capacity of private universities in America (Ginsberg, 2011).

The contribution of soft sciences such as sociology, education and humanities to these figures is very rare. Hard science fields absolutely dominate the global patentable knowledge production and, within this category, the kind of knowledge relevant to commerce and trade. Not only such knowledge is of great interest for the states but, often, they are also state-owned and/or supervised. Furthermore, patent registration processes are monitored by the signatory states of relevant international treaties.

In the typology of knowledge, Lyotard distinguishes the ‘narrative knowledge’ from the ‘scientific knowledge’ (1979/1984). Sociology and humanities pertain to the former kind of knowledge. Contemporary states closely monitor the production of scientific knowledge but, for centuries in the past, a score of imperial states have had strenuous control over either type of knowledge in their colonies.

5.4 Knowledge in Postcolonial Studies

Production of knowledge is one of the most deliberated problems in colonial and postcolonial studies. It is for a reason: Geopolitics surfaces not only etymologically (*vid. supra* ‘*Wissen ist macht*’) but also epistemologically from the modern socio-historical structure of colonial world-system (Mignolo, 2002; Wallerstein, 2004). The following paragraphs illustrate how colonial and postcolonial studies are utmost concerned with knowledge production, that is, on who produces knowledge, what is the nature of its content and from which perspectives such knowledge is generated (e.g., Eurocentrism and Western epistemology).

Frantz Fanon’s *Black Skin White Masks* (1952/1986) triggered the decolonization movement in the 1960s against French racism and the dehumanization of colonial subjects. Fanon called this process ‘authentic disalienation’ (Fanon, 1952/1986, 5), by which locals’ identity, psyche and culture, mediated by language, are to be restored. This insight was further developed in his *The Wretched of the Earth* (Fanon, 1963), which starts with a chapter on the violence as a social, cultural and political currency in colonial territories. Since culture, economy and polity are entangled, violence initiated and instrumentalized by colonizers takes various forms among which cultural discrimination and dehumanization (alienation) outstand. Economic sanctions and tributary impositions are only different modes of the violence. It is within and through the violence that a particular type of new knowledge (language and humanity) with its racial and cultural hierarchical structure, engenders a new human—the colonial subject. This one-way violence does not last forever, however. Fanon draws on his profession of psychiatry to argue that the colonizers’ violence is met with a counter-violence of the colonized in the manner of collective catharsis. Thus, decolonization is not only violent and revolutionary but also a healing process for the colonized.

A half a century later and with many colonial territories freed of physical occupation of their lands by foreign powers, the discourse of imperialization and decolonialization are on the rise in Asia (Chen, 2010; Connell, 2007; Tharoor, 2017; Vukovich, 2013). Tharoor argues, for example, that modern India lost much of its civilization under the British rule, and it achieved independence with only 16% literacy rate, and is still struggling to educate (Tharoor, 2017). He showcases the English language, which was “not a deliberate gift to India, but again an instrument of colonialism, imparted to Indians only to facilitate the tasks of the English...[that is to produce] a class of persons, Indians in blood and colour, but English in taste, in opinions, in morals and in intellect” (2017, p. 122). Colonizers produced deliberately designed history subject for schooling where the superiority of “all things British” was engraved into Indian students. Tharoor and other authors in postcolonial studies commonly argue that schooling curriculum content was at the service of the imperial interests. In fact, knowledge production and reproduction via pedagogy was a tool of colonization.

In contrast, knowledge production appears not as a tool but as the most glaring problem *per se* in Asia according to Chen Kuan-Hsing (2005, 2010). He argues that

knowledge itself is “one of the major sites in which imperialism operates and exercises its power” (Chen, 2010, p. 211). Thus, in the opening of his 2010 opus magnum, Chen calls for “critical intellectuals in the former and current colonies of the third world to once again deepen and widen decolonization movements, especially in the domains of culture, the psyche, and knowledge production” (2010, p. vii).

Decolonization of Asian countries is an incomplete and ongoing process. Decolonization is not about giving up or forgetting colonial experience. Instead, Chen argues, “Asia requires a different sort of knowledge production” (p. 2). Knowledge is not only a problem of history that Asia has to put up with, but a pending homework of Asian societies today. In order to disentangle the *problématique* of colonial knowledge and move on, Chen highlights, there is a need for an alternative knowledge that can prevent “the kind of knowledge work that lays the foundation for future imperial expansion” (p. 34). Chen argues that the logics of market cannot explain well current social and political structure in Asian countries (Chen, 2010, p. 71):

in East Asia, intellectual circles in South Korea, Japan, Hong Kong, and Taiwan cannot use their own existing analytical tools to understand the structural operating logic of the state and society in mainland China, Vietnam, and North Korea. The simple notion of the market does not suffice to explain the social and political formations there. Elements of the socialist era are still operating. To quickly abandon the analytical language of Marxism would not produce a proper understanding of ongoing transformations

Chen considers his intellectual position a “geocolonial historical materialism in order to develop a more adequate understanding of contemporary cultural forms, practices, and institutions in the formerly colonized world” (Chen, 2010, p. 1). This Marxist or neo-Marxist interpretation would constitute a new knowledge of a geopolitical kind, which could overcome the present conditions of knowledge production in Asia. The historical processes of imperialism, colonialism, and the Cold War in Asia were meshed, which in turn, shaped and conditioned the knowledge production phase of GPK. In regard to knowledge produced by the academia, Chen contends that a great number of academics in Asia were trained in the United States and they are “now in power in their countries, where they are implementing another round of modernization. Knowledge production in the region has been heavily influenced by the U.S. academy ever since the end of the Second World War” (Chen, 2010, p. 120). Chen’s concern is not the past but the present and the future of knowledge production.

The main goal of *Asia as method* is, therefore, to transform the current mode of knowledge production so as to halt a mainstream international construct that Asia is at its best an object of analysis. Instead, Chen suggests, Asia should stand tall as a means of transforming knowledge production (Ref. Chen, 2010, pp. 216 & 227) to understand itself and the world at large. Furthermore, Chen does not only suggest challenging the history of imperial/colonial knowledge production. Instead, he suggests an Asian integration to counteract neo-imperial and neo-colonial forces of globalization (Chen, 2010, p. 268):

we cannot allow ourselves to be swept up in the rush toward neoliberal globalization... We have to insist on advancing the critical work of deimperialization, decolonization, and de-cold war, and facilitating regional integration on the level of knowledge production through the practices of Asia as method

At the core of Chen's works lies a strong criticism of knowledge produced by the global power holders. One of its focuses of inquiry is how instrumentalized knowledge serves (neo)imperial and (neo)colonial appetites.

5.5 Decline of Narrative Knowledge

Knowledge production as one of the key problems of colonialism still holds true today, when military might is replaced with neoliberal states' financial, economic and soft power (Nye, 2004). Instead of territorial occupation and exploits, new modes of imperialism and colonialism operate in the current context that the capitalism of old gave way to neoliberal globalization.

Amidst these changes, 'narrative knowledge' (Lyotard, 1979/1984) produced in social sciences and humanities are in steep decline in significance and relevance. In social sciences with cognates such as science of administration, for example, the 'war of epistemologies' has silently, yet effectively replaced the 'war of paradigms' (Park, 2015). With the rise of hard science, theories of knowledge became gradually equated to the philosophy of science, which studies the limits, justification and validity of production of scientific knowledge. Producers of narrative knowledge are fewer and these are overly concerned about epistemology, an inquiry cluster that has been overexploited and ultimately downgraded the relevance of narrative arguments and claims. In my view, to the definition of postmodern "incredulity toward metanarratives" (Lyotard, 1979/1984, p. xxiv), we should now add all sorts of lesser narratives.

The narrative knowledge generated by postcolonial studies have criticized knowledge production by colonial powers and suggest an indigenous knowledge production as an antidote. If instead of their focus on the knowledge production, we were to pick 'knowledge transfer' as the decoder, the postcolonial world as well as neocolonial dynamics of the ongoing globalization with GPK as its motor could be better understood.

5.6 Knowledge Transfer

Almost as soon as knowledge with value is produced, it undergoes a process of appropriation. With this early transfer of ownership, knowledge becomes subject to the interplay between the knowledge producer and acquirer. Knowledge is now in motion. It could be taught, learned, sold, bought, stolen, censored, classified, prescribed or passed on to the public. Let us call this set of possible movements

‘knowledge transfer’ in order to demonstrate that GPK principally hinges around knowledge transfer rather than its production.

Knowledge transfer has been speculated philosophically. Greek metaphysics on *agathos* (good) is the basis of theoretical elaborations on the transferability and the expansive nature of knowledge. Well reflected in the Latin dictum ‘*omne bonum diffusum sui est*’ (every good thing tends to diffuse itself) of Neoplatonists and Neoaristotelians such as Pseudo-Dionysius and Thomas Aquinas (McClymond, 2010, p. 92): Knowledge is a good and, as such, it tends to spread out and disseminate among people.

However, for the same reason that knowledge is a good, its transfer has not always been fluid. On the contrary, knowledge has been associated with intellectual and social hierarchy since the Antiquity as is reflected in the ‘good and wise’ (*agathos kai sophos*) in Plato and ‘superior person’ (君子 *junzi*) in Confucius (Park, 2016). In the Modernity, knowledge has been a key site of international conflicts over resources and market; ideologies with justification of ways of life and values; and, power. These are not mutually exclusive, for example, “power is not only a means for securing economic advantage or ideological maximization, it is also a source of conflict in its own right” (Katz, 1965, p. 374). Knowledge is at the center of contemporary geopolitics with the highest level of tensions in its transfer.

5.6.1 Forced Knowledge Transfer

Teaching and learning are key mechanisms of knowledge transfer. Not seldom, GPK takes the form of top-down knowledge transfer through prescription, coercion and force. The content and mode of transfer of knowledge are the main concern of curriculum and instruction—pedagogy. From the Greek *Paideia* to the Confucian *Great Learning*, the ideal and real goals of pedagogy have been dauntingly broad if not ambiguous.

Today, attainment of knowledge is linked to individual and social development to the highest levels of growth and wealth (Pilling, 2018; Smith, 2016); thus, teaching and learning are depicted as a tool of empowerment or, conversely, disempowerment. Freire’s account of knowledge transfer denounces the problem of pedagogy at the service of power-holders. It underscores a dialectical relation between knowledge taught and received (1972, p. 17):

the radical is never a subjectivist. For him the subjective aspect exists only in relation to the objective aspect (the concrete reality which is the object of his analysis). Subjectivity and objectivity thus join in a dialectical unity producing knowledge in solidarity with action, and vice versa

From the perspectives of oppressor-oppressed dialectics, knowledge flows downward from the objectivity of the oppressor to the subjectivity of the oppressed. There is an upper hand and lower hand in the gradient of power, which Freire connects to his famous *banking* concept: “In the banking concept of education,

knowledge is a gift bestowed by those who consider themselves knowledgeable upon those whom they consider to know nothing” (1972, p. 17). Freire further explains who the protagonists of the referred dialectics are: “the ‘professor’ is the one who has knowledge and to whom they [peasants or oppressed] should listen to. The criteria of knowledge imposed upon them” (ibid. p. 39). The passage also indicates a power hierarchy in knowledge ownership and a direction of flow in knowledge transfer. Freire’s concern is the gap between the oppressor and oppressed, that is, “the distance between the teacher and the taught” (p. 50).

Freire’s account of knowledge transfer draws on the postcolonial state of affairs in Brazil, when citizens were forced to acquire and live with ethics and psychology aligned with the dominant power structure. An ‘educational science’ can be identified with five formal steps in knowledge transfer, say, *à la* Herbart: Preparation, presentation, association, generalization, and application (Hilgenherger, 1993). Earlier in the colonial lands, comparable knowledge transfer was planned and executed. It can be illustrated with a unique case in the history of colonization where several Asian nations were colonized by a neighbor in the same continent. During the Japanese colonial expansion in Asia, the book *Kokutai no Hongi* (trans. Cardinal principles of the national entity of Japan) was painstakingly prepared, one draft after another with obscure authorship and editorship, but under a direct supervision by the *Bureau of Thought Control of the Japanese Ministry of Education* (1937/1949). This extensive and intricate text mythologized and deified the Shōwa Emperor and legitimized his leadership in the geopolitical expansion of the Japanese Empire (Beasley, 1950; Kublin, 1950).

An exorbitant number of *Kokutai no Hongi* copies were printed for colonial subjects and Japanese citizens, and its reading made compulsory in Japanese colonies across Asia (Hyung & Im, 2017). Geopolitical pedagogy of the *Kokutai no Hongi* is not a matter of history. An Osaka kindergarten sponsored by the Japanese Prime Minister Shinzo Abe was reported for its forcing young students to recite the 1890 *Imperial Rescript on Education* by heart and every morning (Sposato, 2018). A passage of the rescript reads, “Should emergency arise, offer yourselves courageously to the State, and thus guard and maintain the prosperity of Our Imperial Throne” (Imperial Rescript on Education, 1890).

Knowledge exchange implies a bartering between two or more parties in comparable footings, hence, it is politically less sensitive than knowledge transfer. For instance, ‘knowledge exchange’ is almost always preferred over the term ‘knowledge transfer’ by universities in their mission statement. It is not clear that the current international relations operate through knowledge exchange. The ongoing neoliberal globalization could hardly be characterized with *dialogics*, which should be in principle a two-way act that characterizes ‘open societies’ (Park, 2009; Popper, 1952). Instead, the default geopolitics appears to be in top-down mode. With G-Summits and trade alliances, globalization enacted by ‘superpower hopefuls’ is but “capital-driven forces [that] seek to penetrate and colonize all spaces on the earth with unchecked freedom, and that in so doing have eroded national frontiers and integrated previously unconnected zones. In this ongoing process of globalization, unequal power relations become intensified, and imperialism expresses itself in a new form” (Chen, 2010, p. 4).

5.6.2 *Knowledge Transfer and Espionage*

Unequal power relations have deepened through the handling of knowledge from colonial era's top-down pedagogy to the current neoliberal globalization. As stakeholders of the global GPK, states openly inculcate knowledge production for profit, for example, the 'Creative Industries' where they encourage enterprises to commodify science-technology, culture and arts for local and international consumption (UNESCO, 2009). Furthermore, states also play a hands-on role in the ownership and management of knowledge. Global patents and copyright regulations are but an example of how capital-oriented and national-interest preoccupied states monitor knowledge transfer.

Since knowledge is a crucial asset and imperative for state power, and not every state can produce as much as it needs; different states react differently to this predicament. As this chapter is written, there are some major geopolitical strains over the transfer of data and information: between the USA and China over trade and military intellectual property; between the USA and Iran-North Korea over nuclear technology; between China and Russia over military aviation, sonar and missile technologies; between China and the West over the information in wireless network; between Hong Kong and the US over personal data storage amidst the National Security Law; and, among all the nations, over combating the COVID-19 pandemic. Furthermore, data breach and cyber-attacks involve almost every country on Earth, hacking being the most prevalent (Verizon, 2020). These cases have in common an unfair or illegal attempt to get hold of knowledge with ensuing reactions. Geopolitics of knowledge occurs at a greater speed and volume than the mediation capacity of international organizations such as the World Trade Organization and the World Health Organization.

For states and industries with capabilities to produce knowledge, research and development is the normal way to sustain growth and keep up with the rate of cutting-edge innovation (Seddighi, 2015). For developing countries and industries, research and development is an expensive and lengthy affair, hence, they look for alternatives such as direct purchase and the importing of knowledge-intensive products. Some states or industries might find it convenient to search for creative ideas from other innovators instead of inventing them on their own, which is an incentive to industrial espionage (Cozzi, 2001). A mega-scale research on the East Germany's state-backed economic espionage demonstrates, for instance, that the industrial and technology gap between East and West Germany was effectively narrowed down through espionage, which strongly suggests that economic and industrial espionage allows, in fact, a quicker access to knowledge with handsome economic returns (Glitz & Meyersson, 2020).

Economic and industrial espionage has its precedence in the eighteenth century industrial espionage between the United Kingdom and France over the coal-fuel technology knowledge transfer to France (Harris, 1998). However, disputes over intellectual possession among the European nations can be further traced back to a Venetian proto-patent law written between 1450 and 1550 (Prager, 1944). History

shows that GPK over knowledge transfer predates the Industrial Era and those countries engaged in such activities often are the “scientifically and industrially weaker party, the party that is learning or trying to catch up” (Søilen, 2016, p. 51). Economic espionage and industrial espionage differ in that the former implies “government’s efforts to collect information, appropriate trade secrets, and steal knowledge” and, the latter, without government involvement (Søilen, 2016, p. 52).

Economic espionage constitutes a key justification of the Sino-American tensions. All American disputes over trade deficit and market access with European and East Asian nations started and ended in March 2018 with the exception of China. By the end of the same month, it was obvious that the real target of the trade dispute was China and not the rest of the American allies such as South Korea and Germany (Rasmus, 2018). The Sino-American trade war started in July 2018 and it has now escalated into diplomatic knock-backs and military’s repositioning. It took little time for political analysts and journalists to label the phenomenon as a new Cold War. Yet, there is a less and a more to it.

Less to it, because China has no ally nation worth its name to form a power bloc, unlike the Soviet-American Cold War when every country in the world took a side or remained non-aligned. China’s lack of ‘friends in need’ is due not only to its poor diplomacy in many territorial and maritime disputes with almost all neighbors but also, above all, due to the enmity brewed by its four decades of self-centered business practices, typically, an international partnership that rather nastily ends in knowledge appropriation and elimination of the partner (e.g., Bloomberg, 2020). More to it, although the Sino-American tensions are ostensible in commerce, their real fault line might follow the differentials in mentality, culture, axiology and practice. The Sino-American tensions as a whole could also be seen as nothing short of a clash of civilizations wherein a civilization is defined as the broadest cultural entity and non-reducible to political entity such as the nation state (Huntington, 1996, p. 312):

A more dangerous source of a global intercivilizational war is the shifting balance of power among civilizations and their core states. If it continues, the rise of China and the increasing assertiveness of this ‘biggest player in the history of man’ will place tremendous stress on international stability in the early twenty-first century.

Regardless of the Huntingtonean foresights, the facts on the ground indicate that Chinese neighbors already seem to feel the tremors of the rise of China (see Russia against China over strategic weapons in Simes, 2019).

In regard to the knowledge transfer aspect of the Sino-American trade war, the ultimate aim of the United States, possibly even after Trump’s tenure in presidency (Kuhn, 2020), is to limit technology transfer, legal or illegal, from the United States to China (Rasmus, 2018). American surveillance on intellectual property theft, both industrial and military, predates the Trump administration. The ‘*U.S.-China Economic and Security Review Commission*’ (USCC) was established by the United States Congress in October 2000 with the legislative mandate to monitor,

investigate, and submit to Congress an annual report on the national security implications of the bilateral trade and economic relationship between the United States and the People's Republic of China, and to provide recommendations, where appropriate, to Congress for legislative and administrative action.

In its hearings and written reports to the US Congress, the USCC uncompromisingly states that China “depends on industrial espionage, forced technology transfers, and piracy and counterfeiting of foreign technology as part of a system of innovation mercantilism” (Munsey, 2013). In a US Department of Justice (FBI) survey of 165 companies, 50% acknowledged that trade secrets or intellectual property had been stolen and 95% suspected China was behind the thefts (Chon, 2015; Dean, 2015). As of July 2019, the FBI had about one thousand investigations into Chinese intellectual property theft, including Beijing's the ‘*Thousand Talents Program*’. This program brings Chinese professionals overseas back to China, in a ‘Brain Gain’ fashion, with huge financial/professional incentives. Yet, to the eyes of the US government, it might be just one more Chinese state machinery that formalizes and exacerbates knowledge theft (AFP, 2019). As argued earlier, the cause of international conflicts is over power, resource and ideology (Katz, 1965) and, to the imperial eyes of the US, arguably, the three causes are: the power of the Chinese Communist Party and the National Science and Technology Leading Group (国家科技领导小组); the resources for the ‘Made in China 2025’ strategy (Tse & Wu, 2018); and, the ideology of ‘The Chinese Dream’ (中国梦).

Instead of producing innovative thinkers, a state, any state, can paradoxically turn inwardly, away from the open of global knowledge flow that is crucial for spearheading authentic innovation. Inward looking strategies in knowledge acquisition leads to a self-imposed ostracism, which in turn, tightens up the control of the media, public education and basic rights such as freedom of expression and intellectual freedom (e.g., academic freedom).

Keeping at bay rivals is neither strange to geopolitics nor illicit, thus, states act upon their contenders with different doses of Machiavellianism. Knowledge transfer through espionage is, let us grant it, a short-term cost-effective means of securing economic development (Glitz & Meyersson, 2020). However, it is also a source of conflict, both internal and external, in its own right. Undermining knowledge constitutes a modality of GPK through a dual maneuver of concealment-exposure through surveillance, censorship, and sabotage.

5.6.3 *Undermining Knowledge*

GPK through surveillance, censorship, and sabotage intensifies in the context of the late modernity with a gargantuan knowledge repository and super-highway of knowledge transfer called the *Internet*. In a world that most of the human and

machine-generated knowledge pass through or are stored in the Internet,² the majority of GPK occur in the cyber-space and through cyber-tools.

Cyber-attack or data breach is a generic name given to illegal incursions into data and information networks regardless of the type of damage inflicted and underlying motives. Before any discussion on surveillance, censorship, and sabotage of knowledge, a phenomenological analysis of concealment and exposure of knowledge is necessary. Concealment is an act of intentionally rendering certain knowledge inaccessible, for example, hiding one's feeling. Contrariwise, specific knowledge can be exposed at convenience, for example, turning their adversaries vulnerable with past scandals.

Neither concealment nor exposure is illicit *per se*. Thomas Nagel (1998) argues that there is a tacit social agreement over making use of concealment to dodge inconvenience or evil: "conventions of reticence and privacy serve a valuable function in keeping us out of each other's faces" (p. 4). At the individual level, concealment of certain knowledge from others can be justified because "the boundary between what we reveal and what we do not, and some control over that boundary, are among the most important attributes of our humanity" (ibid). Furthermore, "concealment includes not only secrecy and deception, but also reticence and non-acknowledgment. There is much more going on inside us all the time than we are willing to express, and civilization would be impossible if we could all read each other's minds" (Nagel, 1998, p. 4). Parallels could be drawn with exposure—it could be for both for advantage or detriment of the involved parties. We can also envisage a significant variation across different civilizations in terms of culturally permissible level of concealment and exposure.

5.6.4 Surveillance

Surveillance rides on knowledge theft and spying but it differs from industrial espionage in the type of knowledge appropriated and, above all, who its legit holder is. The person or group under surveillance is assumed an actual or potential harm inflictor or criminal. The nature of appropriated knowledge could range from personal behavior, strategic information of an inimical state through an entire society or system, such as the 'surveillance society' and 'surveillance capitalism' (Wood, 2006; Zuboff, 2018).

Global mass surveillance via high-speed information and communication technology of the United States has been denounced for its operations without public awareness, let alone consent, against its own federal constitution for that matter, and afflicting the ordinary citizens more than the real criminals (Snowden, 2019). Unlike

²It is impossible to measure the amount of the amassed knowledge in the Internet since knowledge *stricto sensu* should be differentiated from data and information. However, the amount of data can be quantified—the 2018 snapshot and the estimates for 2025 stand respectively in 33 and 175 Zettabytes (1 ZB <>1021 bytes) (Reinsel et al., 2018).

the US that stores non-criminal personal information, the Chinese state gives it a more pragmatic use, say, from facial recognition and automated check-in in airports to color-marked QR Barcode for different level of exposure to the COVID-19. China also uses Big Data for permanent assessment of citizens' behavior and, according to individual points attained, citizens are permitted or banned for trips, their children allowed or denied access to quality education, their real time location made public to phone applications for financial crime prevention and so on—the *Social Credit System* (Kostka, 2019; Song, 2019).

The impact of surveillance on the surveilled is masterfully illustrated by Foucault in his *Discipline and Punish* (1995) with a parallel drawn from Bentham's *Panopticon* prison design. In it, a guard is at the center of a centripetally configured prison building where he can watch every corner of the cells. The greatest effect of such a surveillance, total and continuous, is that the surveilled is in "a state of conscious and permanent visibility that assures the automatic functioning of power (...), the surveillance is permanent in its effects, even if it is discontinuous in its action" (p. 201). More than the kind and amount of information exposed, the main issue of a state-run mass surveillance is its conditioning people's psyche and denigration of freedom. The gravest damage inflicted by state-backed mass surveillance of ordinary people is a devaluation of virtue-oriented axiology of a meritocratic civilization in the name of security and, not without paradox, freedom.

5.6.5 Censorship

Censorship relies on surveillance but its *modus operandi* is not necessarily knowledge theft. Instead, it interferes with knowledge transfer. Censorship occurs in public places and spaces where people are exposed to information/knowledge: Media, books, school textbooks and the Internet. There is an array of modalities, from self-censorship to state-backed Internet firewalls through the cancellation of publication licenses yet, ultimately, all acts of censorship interfere with access to knowledge and freedom of expression.

The dual-right of access to knowledge and freedom of expression was unknown to Asia until the twentieth century. Even in the West, the assurance of not getting beheaded by a monarch after a dissenting speech was legally protected in the seventeenth Century. It is not far fetching that freedom of expression is a cultural and legal breakthrough of civilizations. John Milton, English scholar and poet, in his 1644 *Areopagitica* speech to the Parliament of England opposing licensing and censorship, submitted that freedom of speech is a multi-faceted right, which includes: (1) the right to seek information and ideas; (2) right to receive information and ideas; and, (3) right to impart information and ideas (Milton & Sabine, 1951). According to George Sabine, the key principle of *Areopagitica* was the "right and also the duty of every intelligent man to know the grounds and take responsibility for his beliefs and actions" (Milton & Sabine, 1951, p. ix). Milton's point was that, if a text is to be rejected, it should first be examined, refuted, and condemned rather

than prohibited before its ideas have even been expressed (Milton & Sabine, 1951). Milton did not succeed in convincing his parliament. Yet, the 1688 *Bill of Rights*, one of the basic documents of English constitutional law, documented the freedom of speech in parliament, and it is still in effect in the form of the modern parliamentary privilege (Bill of Rights, 1688).

Back to the problem of censorship, in the Internet in particular, we need to define what we mean by *Net Neutrality*. The debate over net neutrality originated in the capitalist world when *Internet Service Providers* (ISPs) started juggling new business strategies to generate income, for instance, pay-to-access the Internet, section or the whole, and pay-for-content (Krämer et al., 2013). Net neutrality is a multifaceted discourse in defense for the Internet to remain accessible, open and free to the public. A perfect net neutrality is, therefore, an ideal because after years of debate, people still pay ISPs to use optical cables and servers, buy content, and pay service commission in retail hardware purchase. But, then, we have the autonomy of the state-run cyber-censorship.

The *Great Firewall* paired with the *Big Data* surveillance is one of, if not the most successful censorship system that controls access to knowledge and freedom of expression over the Internet. Apart from the concealment of a big chunk of foreign information, its success also has to do with exposure of individual particulars, for example, compulsory user registration with real identity. A common mode of this censorship is tracking politically sensitive ideas and opinion in the social media, and even deleting them with no warning. China recognizes no net neutrality since this ideal is at odds with everything that the Chinese state is trying to do in terms of control over its population and their knowledge. The *Great Firewall* is filtering the wealth of information on the Internet and has a significant impact on the information available to citizens, in particular, knowledge-intensive professionals. University academics in mainland China cannot access a wide range of academic papers in social sciences and humanities. The reduction of the variety of information and knowledge would ultimately disempower people to be innovative and competitive at the global stage.

A precedence in Asia was the isolation period of the Tokugawa Japan (1639–1853). While isolation allowed Japan to unite inwardly, they ended up woefully behind technological and scientific knowledge of the West. The ensuing Meiji restoration pushed Japan to imitate European geopolitics of colonialism cum militarism, which eventually ruined them. In striking contrast with Den Xiaoping's doctrine of openness and low profile (Wang, 2014), China has lately been regressing into self-isolation with undiplomatic assertiveness, inward and outward censorship, and aggressive territorial disputes. The more China self-ostracize, the more it will engage in illegal knowledge transfer.

5.6.6 Sabotage

Another category of GPK is *sabotage*, that is, “any disabling damage deliberately inflicted, especially that carried out clandestinely in order to disrupt the economic or military resources” (OED, 2020). The goal of knowledge sabotage is neither spying nor getting hold of valuable/useful information. Instead, it aims to destroy knowledge held by power adversaries and ideological antagonists.

Sabotage is the most likely method to undermine knowledge and information in international confrontations. Future military confrontations and battlefields will mostly be online; its casualties counted in Gigabytes-cash; and, its main weapon, hacking. In this scenario, the likely victor would be China by the sheer number of hackers and number of cyber-attacks. China is today the largest cyber-attacking country with about 27% of all cyber-attacks directly from China and up to 41% if redirected attacks were counted (Baig, 2017). China followed by the US with about 17%, Turkey 10%, Brazil 8% and Russia 5% with the American cyber-attacks receiving an honorable mention for sophistication (Baig, 2017).

Finally, a new front of international sabotage of knowledge is of an anarchic kind. Hedonistic motivation increasingly underlies many acts of sabotage of knowledge as many perpetrators inflict damage out of pleasure, thrill, catharsis or self-assertiveness.

5.7 Discussion and Conclusion

We are experiencing a sizable de-globalization effect of the COVID-19 pandemic with the rise of a distilled form of homophobia, that is, the fear of human beings for sheer fact of being human. The pandemic also reverberates long-standing injustice such as racism, gender bias and core-periphery marginalization. From the economics of lockdown to vaccine technology, knowledge has proven to be, once more, the core of the pandemic-related challenges. Geopolitics of knowledge is real, intense and of the highest relevance because of the social imagery and discourse of knowledge economy meshed with the imperatives of knowledge as power and the dread of relegation to the wrong side of the master-slave dialectics (Hegel & Baillie, 1807/1949).

A differential analysis of knowledge production and knowledge transfer—two faces—provides important insights into global geopolitics. About the knowledge production, we witness a significant decline in the production of ‘narrative knowledge’ (Lyotard, 1979/1984) but not necessarily due to a postmodern incredulity toward metanarratives (Park, 2018). Instead, the decline of ‘narrative knowledge’ might have more to do with the rise of commercially valuable ‘scientific knowledge’ (Lyotard, 1979/1984) amidst globalization, and, hence, sponsored and promoted by states and industries.

As it transpires in the field of postcolonial studies, scholars in political philosophy/science usually pay attention to the production process of knowledge and its essential characteristics. I submit that instead of such a static state of knowledge, the inquiry into geopolitics should pay more attention to the phenomena of ‘knowledge in motion’ such as transfer, dissemination, pedagogy, indoctrination, theft, espionage, surveillance and censorship.

There is neither better nor more current example of geopolitical dynamics of knowledge than the ongoing Sino-American tensions. We witness a quickly escalating *decoupling*³ in trade, technology, diplomacy, military, and the Internet. The strongest and the most consistent *casus belli* of this transpacific ‘Cold War II’ is a systematic and persistent knowledge theft and espionage. This *casus belli* is, of course, from the perspective of the hegemonic discourse of the *America First*, which is manifestly different from Chinese civilization’s emperor-subject hierarchy, power legitimization construct, and the idea of what knowledge is and should be for no less hegemonic discourse of *The Chinese Dream*. In the Chinese civilization, knowledge transfer has usually been speculated upon from the perspective of an instructor-learner hierarchy and the moral excellence of the learner (Park, 2016), while it assigns little or no weight to the principles of honoring contracts, separation of powers, rights and duties.

However, there is also a sizable similarity between the two civilizations entangled in the Sino-American geopolitics of knowledge—an avidity for wider recognition and legitimation of power. Jean-François Lyotard foretold that the main problem of knowledge in the late (post)modernity is the *legitimation by power* in the post-industrial and technologically able societies (1979/1984, p. 47):

This is how legitimation by power takes shape. Power is not only good performativity, but also effective verification and good verdicts. It legitimates science and the law on the basis of their efficiency, (...) thus the growth of power, and its self-legitimation, are now taking the route of data storage and accessibility, and the operativity of information.

A decade later, towards the end of his career, Lyotard (1988/1991) argued that the discourse of development is the dominant ideology of our time. ‘Development’ is a *quasi-metaphysics* that needs no finality and it is “not attached to an Idea, like that of the emancipation of reason and of human freedoms” (p. 7), hence, the metaphysics of development is rather inhuman. Scientific and technological knowledge as well as philosophical knowledge only substantiates the ‘discourse of development, which is used in turn by those in authority and power to legitimate their efficiency. Technology and the new media exert influence on the discourse of development through delocalization and detemporalization of cultures, hence, technological revolutions allow industries to spread across civilizations (Lyotard, 1988/1991). However, in geopolitics, technological revolutions also play a subversive role, that of undermining, sabotaging and pillaging industries, knowledge and power.

³Decoupling: “Separation of previously linked systems so that they may operate independently” (CED, 2012)

The subversive power of technology-mediated knowledge is accurately reflected in the momentous development and enthusiasm for the *Big Data*. Beyond a mere repository of mega-data, Big Data entails people's capabilities to transform data corpus into information and knowledge, then mobilize it purposefully. In research, with the analysis of large-scale data with *ad hoc* statistical models such as multi-level analysis, path analysis and latent cluster analysis, Big Data is a robust inductive instrument to validate researchers' hypothesis (Park, 2019). Big Data legitimates scientific knowledge as Lyotard argued (1979/1984). However, in a double-edge sword fashion, Big Data is also used for the legitimation of geopolitics at the service of concealing and exposing specific knowledge at the convenience of power holders.

"The sovereignty lieth hid in knowledge" remarked Francis Bacon in the early modern globalization (1597/1825, p. 255). Today, as it was during his time, sovereignty of individuals, groups and nation-states lies in the capacity and ability in the production and administration of knowledge. Knowledge has been not only power but also the main currency of civilizations and epochs. Knowledge continues to be both the goal and the means to hold and sustain power in global geopolitics. *Geopolitics of knowledge* is, thus, the political dynamics for knowledge and through knowledge among people in different geophysical places and discursive spaces.

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